COURSE: Cyber Crimes, Forensics, and Incident Management

Assignment: Literature Review Paper - Deadline: 31st October 2026

Total Marks - 20

Objective

The purpose of this assignment is to develop a comprehensive understanding of cyber-crimes, digital forensics, and incident management by critically analyzing existing scholarly literature. Students will explore current research, identify gaps, and synthesize findings to highlight best practices and future directions in the field.

Instructions

1. Paper Requirements

- **Length:** 4000–5000 words (excluding references and appendices)
- **Format:**
  - Times New Roman, 12-point font
  - Double-spaced
  - APA/IEEE citation style

2. Sections to Include

Abstract

- A concise summary (200–250 words).
- Should briefly state the purpose, scope, key themes (cyber-crimes, forensics, incident management), and main findings.
- Written after completing the paper, but placed at the beginning.

Introduction

- Define cyber-crimes, digital forensics, and incident management.
- State the scope and purpose of the review.

a) Cyber Crimes

- Types of cyber-crimes (phishing, ransomware, identity theft, cyber terrorism, etc.).
- Trends and statistics from recent studies.
- Impact on individuals, organizations, and governments.

b) Digital Forensics

- Role of forensics in investigating cyber-crimes.
- Tools, techniques, and methodologies (e.g., disk forensics, network forensics, mobile forensics).
- Challenges in evidence collection and admissibility in court.

c) Incident Management

- Incident response lifecycle (Preparation, Detection, Containment, Eradication, Recovery, Lessons Learned).
- Standards and frameworks (e.g., NIST, ISO/IEC 27035).
- Case studies of effective incident management.

Critical Analysis (for a or b or c – any one topic)

- Compare and contrast different approaches.
- Identify gaps in current research.
- Discuss emerging technologies (AI, ML, blockchain, cloud security).

Conclusion

- Summarize key findings.
- Suggest future research directions.

References

- Minimum of 25-35 scholarly sources (journals, conference papers, books) I APA style.

3. Evaluation Criteria

- **Depth of Research (30%)** – Use of credible, scholarly sources.
- **Critical Thinking (30%)** – Ability to analyze, compare, and synthesize literature.
- **Organization & Clarity (20%)** – Logical flow, clear writing, and proper formatting.
- **Referencing (20%)** – Proper citation style and avoidance of plagiarism.

4. Submission Guidelines

- Submit via the Taxila portal in PDF/Word format with plagiarism report attached at the end of the paper.

Deadline: 31st October 2026

- Late submissions will incur grade penalties unless prior approval is obtained.

All the Best !!

Following illustrative topics are deliberately longer, more descriptive, and academically styled, making them suitable for reputable journals that expect clarity, specificity, and scholarly framing:

Cybercrime & Criminal Behavior

1. From Traditional Computer Crime to Sophisticated Cyber Offenses: A Comprehensive Literature Review
2. Typologies of Cyber Hackers: Motivations, Profiles, and Behavioral Patterns in Digital Crime
3. Cyberspace as a Criminological Domain: Theories of Criminal Behavior in Virtual Environments
4. The Evolution of Cybercrime Legislation: Assessing the Role of the Information Technology Act
5. Global Perspectives on the Changing Landscape of Cybercrime: Emerging Threats and Policy Responses

Digital Forensics Foundations

6. Networking Fundamentals in Digital Forensics: A Review of Core Concepts and Applications
7. Comparative Forensic Approaches to Windows, Linux, and DOS Systems: Challenges and Opportunities
8. The Emergence of Digital Forensics as a Scientific Discipline: Historical and Contemporary Perspectives
9. Professionalization of Digital Forensics: Skills, Standards, and Ethical Considerations
10. Techniques and Methodologies in Digital Forensics: A Systematic Review of Current Practices

Identity Theft & Computer Crime

11. Identity Theft and Fraud in the Digital Era: Forensic Challenges and Investigative Strategies
12. Traditional vs. Contemporary Computer Crime: A Literature Review of Shifting Paradigms
13. Securing and Preserving Digital Evidence: Best Practices in Cybercrime Investigations
14. Digital Hashing and Evidence Integrity: A Critical Review of Validation Techniques
15. Case Studies in Digital Evidence Seizure: Lessons Learned from High-Tech Investigations

⚙ Forensic Process & Validation

16. Phases of the Digital Forensic Process: A Review of Models and Methodologies
17. Comparative Analysis of Digital Forensic Process Models: Strengths, Weaknesses, and Applications
18. Ensuring Evidence Integrity: Literature Review on Digital Forensics Analysis and Validation

💾 Disk Structures & Data Hiding

19. File Systems and Disk Structures in Digital Forensics: A Review of Acquisition Techniques
20. RAID Data Acquisition in Forensic Investigations: Challenges and Emerging Solutions
21. Steganography, Encryption, and Hidden Partitions: Data-Hiding Techniques in Cybercrime
22. Password Recovery in Digital Investigations: Tools, Techniques, and Ethical Considerations
23. Forensic Recovery of Graphics Files: Literature Review of Methods and Applications

🌐 Network, Cloud & Mobile Forensics

24. Network Forensics in Cybercrime Investigations: Techniques for Traffic Analysis and Attack Attribution
25. Cloud Forensics in Virtualized Environments: Challenges of Evidence Collection and Preservation
26. Honeypots in Cybersecurity: Their Role in Digital Forensics and Threat Intelligence
27. Mobile Device Forensics: SIM Card Structures, Evidence Extraction, and Investigative Challenges
28. Investigating Web and Wireless Attacks: A Literature Review of Forensic Approaches

🛠 Tools & Labs

29. Digital Forensics Hardware and Software Tools: A Systematic Review of Capabilities and Limitations
30. Evaluating Digital Forensics Tools: Criteria, Effectiveness, and Case Applications
31. Accreditation Standards for Digital Forensics Laboratories: Global Perspectives and Challenges

32. Designing Forensic Laboratories: Physical Infrastructure Requirements and Best Practices

🏢 Organizations, Criminology & Organized Crime

33. Organizational Involvement in Cybercrime: A Literature Review of Corporate and State Actors
34. Criminological Theories Applied to Cybercrime: Insights from Routine Activity and Strain Theory
35. Organized Crime and Technology: The Convergence of Traditional Syndicates and Digital Networks

📧 Internet & Email Crime

36. Investigating Internet Crime: Methodologies, Case Studies, and Emerging Challenges
37. Forensic Approaches to Internet Crime Scene Investigation: Tools and Techniques
38. Email Forensics: Examining Headers, Tracking Techniques, and Investigative Challenges
39. Case Studies in Email Crime Investigations: Lessons Learned and Best Practices
40. Remediation in Internet Crime Cases: A Literature Review of Strategies and Outcomes

🔐 Enterprise Security & Infrastructure

41. Vulnerabilities in Computer Network Infrastructure: Forensic Perspectives on Detection and Response
42. Enterprise Security Attacks: Challenges, Trends, and Countermeasures in Digital Forensics
43. Information Security Protocols and Best Practices: A Literature Review for Incident Handling

🚨 Incident Detection & Response

44. Pre-Incident Preparation in Cybersecurity: Building Resilient Systems through Forensic Readiness
45. Incident Detection and Characterization: Literature Review of Techniques and Case Studies
46. Incident Response Processes: Tools, Techniques, and Best Practices in Digital Forensics
47. Investigating Applications in Incident Response: A Systematic Review of Forensic Approaches

48. Case Studies in Incident Remediation: Lessons Learned from High-Tech Investigations

Report Writing & Documentation

49. Standards and Guidelines for Digital Forensic Report Writing: A Literature Review
50. Report Writing in High-Tech Investigations: Challenges, Best Practices, and Emerging Tools

Emerging Trends & Miscellaneous Topics

51. Emerging Cybercrime Trends: Future Challenges, Concerns, and Policy Implications
52. Cybercrime in the Indian Context: Issues, Recommendations, and Legislative Perspectives
53. Internet Governance Models: A Literature Review of Global Approaches and Challenges
54. Data Protection Laws in the Age of Big Data and AI: Forensic and Legal Perspectives
55. Malicious Cyber Activity: Distribution, Attribution, and Jurisdictional Challenges
56. Balancing Information Privacy and Law Enforcement: A Literature Review of Legal Frameworks